

Explicit Viability Bounds for Safety Filters with Configuration-Dependent Braking Authority

Hadi Hajieghrary¹ and Paul Schmitt²

Abstract—A cable-suspended payload may approach a wall while every cable pulls it toward the wall. Although a high-order control barrier function (HOCBF) can admit an input now, the filter can lose feasibility before a quadrotor has time to swing behind the payload. We give a conditional, open-set certificate for this failure and a two-quadrotor example. A thrust-reserved maneuver yields a stopping-distance barrier whose safe set is controlled invariant for the continuous-time taut-cable model. An optimistic stopping distance bounds the viability kernel from outside. Both distances use the cable-direction-dependent support function of a tension zonotope. Simulations expose substantial conservatism when swinging is required, and wall contacts under a sampled filter and slack-cable model. These results distinguish the continuous-time certificate from its unverified sampled implementation.

I. INTRODUCTION

A quadrotor can pull a cable-suspended payload but cannot push it. If all vehicles are between a moving payload and a wall, changing the tensions cannot brake the payload; a vehicle first has to swing to its far side. This delay is hidden by a high-order control barrier function (HOCBF) that tests only the currently achievable acceleration. We quantify the delay and construct a safety certificate from the maximum wallward excursion of a feasible braking maneuver.

Input-constrained HOCBFs [1], [2] and backup CBFs [3], [4] address feasibility through different safe sets. Cable-robot wrench calculations describe forces available at a fixed geometry [5], [6]; aerial transport models describe how that geometry moves [7], [8]. Here the cable directions determine a time-varying tension-force zonotope, and we use a realizable swing-and-brake maneuver to construct an explicit inner bound and a relaxed swing model for an outer bound on viability. Closed-form backup control is also studied in [4]; our closed form concerns the future wall excursion for this particular authority model.

We let each vehicle reserve part of its thrust for the cable tension and part for swinging the cable; the braking authority that this guarantees depends on the cable directions alone (Sec. IV). When every cable swings to its limit in minimum time, the payload covers at most the stopping distance D toward the wall, which is computed exactly (Sec. IV-B). The authority barrier is $H := h - D$: the states with $H \geq 0$ and admissible swing states form a controlled-invariant set, on which the filter of Sec. V always has an admissible input. With upper bounds on the authority and on the swing acceleration,

¹Hadi Hajieghrary (e-mail: hadi.hajieghrary@gatech.edu).

²Paul Schmitt (e-mail: pauls@massrobotics.org, paul.schmitt@reynolds-moore.com).

The authors prepared this manuscript and present it solely in their individual capacities.

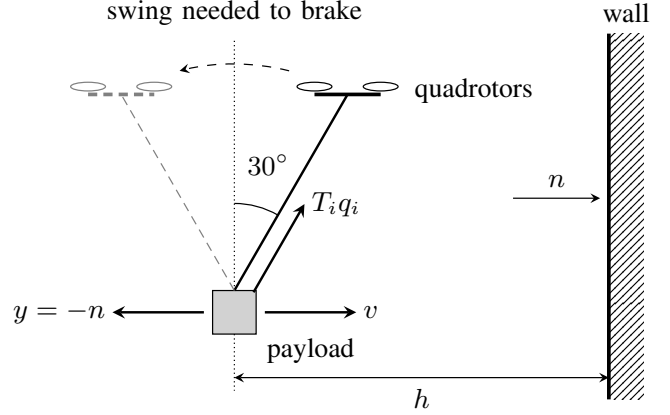


Fig. 1. Two quadrotors carry a payload toward a wall. With both cables leaning toward the wall, positive tensions cannot brake it; a cable must first swing to the far side (dashed).

the same calculation gives an optimistic stopping distance D^{rel} , below which no state is viable (Theorem 9). Both bounds hold for every system whose braking authority is bounded by nondecreasing functions of coordinates that the inputs drive only through their second derivative (Sec. IV-A); the cable system is one instance.

The contributions are: (i) a precise condition under which an initially feasible HOCBF filter must lose feasibility before a cable reaches the braking half-space; (ii) a continuous-time, input-constrained authority barrier and an outer viability bound; and (iii) simulations that measure both the bound's conservatism and the substantial failure of a direct sampled implementation. The latter concerns the implementation and gives no safety guarantee. In simulation, the HOCBF filter lost feasibility in all 690 trials, and the sampled authority filter reached the wall in 59 of 100 trials started near the boundary of its safe set.

For the two-vehicle example in Fig. 1, $m_L = m_i = l_i = 1$ in SI units, each cable leans 30° toward the wall, and $T_{\min} = 1$ N. At $h = 1$ m and $v = 0.85$ m/s, a linear HOCBF with gains $(1, 20) \text{ s}^{-1}$ admits the minimum tensions, but its safe-set condition tightens before a cable can reach the vertical at the allowed 2 rad/s swing rate. Section III gives the exact conditional statement.

II. PROBLEM FORMULATION

Quadrotor $i \in \{1, \dots, N\}$ has position $p_i \in \mathbb{R}^3$, mass m_i , and thrust vector $u_i := f_i R_i e_3$, where $f_i \geq 0$ is the thrust magnitude, $R_i \in SO(3)$ the attitude, and e_3 the vertical unit vector; an attitude controller produces the commanded thrust vectors (Remark 14). Each thrust vector lies in the ball

$\mathcal{F}_i := \{u \in \mathbb{R}^3 : \|u\| \leq f_{\max,i}\}$; a tilt limit can be added to $\mathcal{F}_i$ as long as Assumption 4 is checked with it.

The payload is a point mass m_L at $x_L \in \mathbb{R}^3$, attached to quadrotor i by a massless, inextensible cable of length l_i with unit direction $q_i \in \mathbb{S}^2$ from the payload to the quadrotor, $p_i = x_L + l_i q_i$. The cable exerts the force $T_i q_i$ on the payload and $-T_i q_i$ on the quadrotor, with $T_i \geq 0$ because a cable cannot push. With disturbance forces d_L , d_i and the gravitational acceleration g ,

$$m_L \ddot{x}_L = \sum_i T_i q_i - m_L g e_3 + d_L, \quad (1)$$

$$m_i \ddot{p}_i = u_i - m_i g e_3 - T_i q_i + d_i. \quad (2)$$

The state is $x := (x_L, \dot{x}_L, q, \dot{q})$ with $q := (q_1, \dots, q_N)$. In this work, we assume that every cable is taut, with a tension of at least $T_{\min} > 0$; the filter of Sec. V enforces this bound, and the simulations model slack cables.

We decompose each thrust vector along the cable and perpendicular to it, $u_i = s_i q_i + u_i^\perp$ with $s_i := q_i^\top u_i$, $u_i^\perp := P_i u_i$, $P_i := I - q_i q_i^\top$, and we use the *specific force* of the payload, $a := \ddot{x}_L + g e_3$, which an accelerometer on the payload measures. Differentiating $\|p_i - x_L\| = l_i$ twice and substituting (1)–(2) gives the following.

Lemma 1 (Structure). *Along any solution of (1)–(2) with taut cables: (a) $T_i = s_i - m_i q_i^\top a + m_i l_i \|\dot{q}_i\|^2 + q_i^\top d_i$, so the tension depends on the inputs only through the parallel components $s = (s_1, \dots, s_N)$; (b) $\ddot{q}_i = \frac{1}{m_i l_i} u_i^\perp - \frac{1}{l_i} P_i a - \|\dot{q}_i\|^2 q_i + \frac{1}{m_i l_i} P_i d_i$, so the cable direction depends on the inputs only through $u_i^\perp$ and a ; (c) $M_a(q) a = \sum_i s_i q_i + \sum_i m_i l_i \|\dot{q}_i\|^2 q_i + \sum_i (q_i^\top d_i) q_i + d_L$ with $M_a(q) := m_L I + \sum_i m_i q_i q_i^\top \succeq m_L I$, so a is affine in s ; (d) for a given a , (a) and (b) involve only the quantities of quadrotor i .*

Proof scheme. Differentiate $p_i = x_L + l_i q_i$ twice. Since $q_i^\top \dot{q}_i = 0$ and $q_i^\top \ddot{q}_i = -\|\dot{q}_i\|^2$, projecting the vehicle equation along q_i isolates T_i and gives (a). Projecting perpendicular to q_i removes the tension and gives (b). Substituting (a) into the payload equation gives (c). For any nonzero r , $r^\top M_a r = m_L \|r\|^2 + \sum_i m_i (q_i^\top r)^2 > 0$, so a is uniquely determined by the parallel thrusts. The remaining local dependences in (d) follow directly. $\square$

In words, the parallel component sets the tension, the perpendicular component swings the cable, the two share the limit $\|s_i q_i + u_i^\perp\| \leq f_{\max,i}$, and the quadrotors interact only through the payload's specific force. For fixed $(q, \dot{q})$, (a) and (c) define an affine bijection between s and $T = (T_1, \dots, T_N)$, so we use the tensions as decision variables: given T , $a = \frac{1}{m_L} (\sum_j T_j q_j + d_L)$, and (a) returns $s_i(T)$.

A. Wall constraint, safe set, and admissible inputs

We consider a wall with horizontal unit normal n pointing into it, with the constraint

$$h(x_L) := d_0 - n^\top x_L \geq 0, \quad (3)$$

where h is the distance from the payload to the wall. We write $v := n^\top \dot{x}_L$ for the approach speed, $y := -n$ for the braking direction, and $b := y^\top \ddot{x}_L$ for the braking deceleration,

so that $\dot{h} = -v$ and $\dot{v} = -b$, and we describe each cable direction by $z_i := q_i^\top y$ and $w_i := q_i^\top r$, $r := e_3 \times y$. A cable with $z_i > 0$ leans away from the wall and pulls the payload away from it; one with $z_i < 0$ pulls it toward the wall.

We impose the operational constraints

$$\mathcal{X}_{\text{op}} := \{x : |z_i| \leq \bar{z}, (w_i, \dot{w}_i) \in \mathcal{V}(\nu_w, \bar{w}), \|\dot{q}_i\| \leq \bar{\omega} \forall i\}, \quad (4)$$

with $\bar{z}^2 + \bar{w}^2 \leq \sin^2 \theta_q$, $\theta_q < \pi/2$, which keep every cable within θ_q of the vertical, reserve stopping room for the transverse swing, and bound its swing rate; $\mathcal{V}$ is defined in (10) below. The safe set is $\mathcal{C} := \{x : h \geq 0\} \cap \mathcal{X}_{\text{op}}$. The inputs must respect the thrust limits, the tension bound, and a bound $a_{\max}$ on the payload's specific force:

$$\mathcal{U}(x) := \{u : u_i \in \mathcal{F}_i, T_i(x, u) \geq T_{\min} \forall i, \|\sum_i T_i(x, u) q_i\| \leq m_L a_{\max}\}. \quad (5)$$

The last constraint is a design limit on the payload's acceleration; the bound that follows from the thrust limits alone, $\frac{1}{m_L} \sum_j (f_{\max,j} + m_j l_j \bar{\omega}^2)$, is too large for Assumption 4 to hold with realistic constants.

Tensions affect $\ddot{h}$ directly, whereas a transverse input first changes $\ddot{q}_i$ and thereby later changes the feasible braking force.

A state is *viable* if some input signal with values in $\mathcal{U}(x)$ keeps the trajectory in $\mathcal{C}$ forever; the set of viable states is the *viability kernel* $\text{Viab}(\mathcal{C})$ [9], outside of which no controller can keep the system safe. A set is *controlled invariant* if some admissible input signal keeps every trajectory that starts in it inside it; every controlled-invariant subset of $\mathcal{C}$ lies in $\text{Viab}(\mathcal{C})$, which has no closed form. The problem is: (a) find a controlled-invariant set $S \subseteq \mathcal{C}$ whose membership can be tested online; (b) find an explicit set that contains $\text{Viab}(\mathcal{C})$; and (c) derive a safety filter that is feasible at every state of S .

III. LOSS OF FEASIBILITY OF HIGH-ORDER CBF FILTERS

For a unit vector y , the *exact directional authority* at x ,

$$\alpha_y^{\text{ex}}(x) := \max\{y^\top \ddot{x}_L(x, u) : u \in \mathcal{U}(x)\}, \quad (6)$$

is the largest acceleration that admissible inputs can give the payload along y ; the maximum exists because $\ddot{x}_L$ is affine in u and $\mathcal{U}(x)$ is compact and convex ($-\infty$ if empty). Since it is attained, for any function β of the state,

$$\{u \in \mathcal{U}(x) : b(x, u) \geq \beta(x)\} \neq \emptyset \iff \alpha_y^{\text{ex}}(x) \geq \beta(x). \quad (7)$$

Since h has relative degree two with respect to the tensions, HOCBFs [10], [11] apply the barrier condition twice. Let α_1 be a continuously differentiable class- $\mathcal{K}$ function, α_2 an extended class- $\mathcal{K}$ function, and $\psi_1 := \dot{h} + \alpha_1(h) = -v + \alpha_1(h)$. With $\dot{v} = -b$, the HOCBF condition $\dot{\psi}_1 \geq -\alpha_2(\psi_1)$ reads

$$b \geq \beta(x) := \alpha_1'(h) v - \alpha_2(\psi_1). \quad (8)$$

The HOCBF filter minimizes the change to the nominal command over $K_{\text{HO}}(x) := \{u \in \mathcal{U}(x) : b(x, u) \geq \beta(x)\}$ and, while $K_{\text{HO}} \neq \emptyset$, keeps the state in $\mathcal{C}_{\text{HO}} := \{h \geq$

0, $\psi_1 \geq 0$ [1]. By (7), $K_{\text{HO}}(x) \neq \emptyset$ if and only if the feasibility margin $\mu(x) := \alpha_y^{\text{ex}}(x) - \beta(x)$ is nonnegative.

Definition 2 (No-braking set). *The no-braking set for the braking direction y is $\Sigma_y := \{q \in (\mathbb{S}^2)^N : q_i^\top y \leq 0, i = 1, \dots, N\}$.*

In Σ_y no cable leans away from the wall, so no admissible input can decelerate the payload; the set has nonempty interior and contains the configurations of a team accelerating toward the wall.

Theorem 3 (Conditional loss of HOCBF feasibility). *Assume zero disturbances and a feasible interior state $x_0 \in \mathcal{C}_{\text{HO}} \cap \mathcal{X}_{\text{op}}$ with $q_i^\top y < 0$, $0 < v_0 < \alpha_1(h_0)$, and $\mu(x_0) > 0$. Let $\tau = \min_i (-z_i(0))/\bar{\omega}$ and $t_\psi = [h_0 - \alpha_1^{-1}(v_0)]/v_0$. If $t_\psi < \tau$, then every admissible trajectory that remains in $\mathcal{X}_{\text{op}}$ has a nonempty interval before τ with $h > 0$, $q \in \Sigma_y$, and $K_{\text{HO}} = \emptyset$. The property persists on an open neighborhood of x_0 if the input set varies continuously and stays nonempty there.*

Proof. Until τ , $z_i(t) \leq z_i(0) + \bar{\omega}t \leq 0$. Because $T_i \geq T_{\min}$, $b = m_L^{-1} \sum_i T_i z_i \leq 0$, hence $v(t) \geq v_0$ and $h(t) \leq h_0 - v_0 t$. The first time h reaches $\alpha_1^{-1}(v_0) > 0$ occurs by $t_\psi < \tau$. Immediately afterwards $\psi_1 < 0$ while $h > 0$. Since $\alpha'_1(h) \geq 0$ and α_2 is extended class- $\mathcal{K}$, $\mu = \alpha_y^{\text{ex}} - \alpha'_1(h)v + \alpha_2(\psi_1) < 0$. The inequalities are strict at the initial state and the authority maximum is continuous under the stated regularity, so they persist locally. $\square$

The initial-feasibility hypothesis cannot be removed. For example, as $\psi_1 \downarrow 0$ with $\alpha_1(h) = kh$ and $\alpha_2(s) = \epsilon s$, the required braking tends to $kv > 0$, whereas the available braking in Σ_y is nonpositive; such initial states already have an infeasible QP. Thus a universal claim for every choice of HOCBF gains would be too strong. For $k_1 = 1$, $k_2 = 20 \text{ s}^{-1}$, $h_0 = 1 \text{ m}$, $v_0 = 0.85 \text{ m/s}$, and two 30° cables, the floor tensions give $b = -1 \text{ m/s}^2$, the HOCBF threshold is -2.15 m/s^2 , $t_\psi = 0.176 \text{ s}$, and $\tau = 0.25 \text{ s}$ when $\bar{\omega} = 2 \text{ rad/s}$; provided those tensions satisfy the thrust constraints, this is an initially feasible instance.

IV. THE AUTHORITY BARRIER

To obtain an authority that depends only on the cable directions, we fix in advance how each quadrotor divides its thrust between tension and swinging. Throughout, $(z)_+ := \max\{0, z\}$ and $(z)_- := \max\{0, -z\}$.

Assumption 4 (Thrust reserve). *There exist constants $\bar{T}_i \geq T_{\min}$ (tension limit) and $\rho_i \geq 0$ (swing reserve) with $\sum_j \bar{T}_j \leq m_L a_{\max}$ such that, for every $x \in \mathcal{X}_{\text{op}}$, every T with $T_{\min} \leq T_j \leq \bar{T}_j$ for all j , and every $u_i^\perp \perp q_i$ with $\|u_i^\perp\| \leq \rho_i$, the thrust vector $s_i(T)q_i + u_i^\perp$ lies in $\mathcal{F}_i$.*

In words, each quadrotor can produce any tension up to $\bar{T}_i$ and, simultaneously, any perpendicular thrust up to ρ_i , and such tensions respect the cone in (5); by Lemma 1(a), the assumption holds if $(\bar{T}_i + m_i a_{\max} + m_i l_i \bar{\omega}^2)^2 + \rho_i^2 \leq f_{\max, i}^2$.

Definition 5 (Directional authority). *For a unit vector y and $q \in (\mathbb{S}^2)^N$, let $\phi_i(z) := \bar{T}_i(z)_+ - T_{\min}(z)_-$. The directional authority of q along y is*

$$\alpha_y(q) := \frac{1}{m_L} \sum_{i=1}^N \phi_i(q_i^\top y) - g e_3^\top y. \quad (9)$$

A cable that leans along y contributes most at $\bar{T}_i$, a cable that leans against y opposes least at $T_{\min}$, and the gravity term vanishes for the horizontal braking direction; each ϕ_i is nondecreasing, convex, and piecewise linear with one breakpoint.

Lemma 6 (Authority as a support function). *Let $\mathcal{Z}(q) := \{\sum_i T_i q_i : T_{\min} \leq T_i \leq \bar{T}_i\}$. Then $\mathcal{Z}(q)$ is a zonotope, and $\alpha_y(q) = \frac{1}{m_L} \sigma_{\mathcal{Z}(q)}(y) - g e_3^\top y$, where $\sigma_{\mathcal{Z}}(y) := \max_{f \in \mathcal{Z}} y^\top f$ is its support function. Under Assumption 4 and without disturbances, every value of $y^\top \ddot{x}_L$ in $[-\alpha_{-y}(q), \alpha_y(q)]$ is produced by some input in $\mathcal{U}(x)$ at every $x \in \mathcal{X}_{\text{op}}$, so $\alpha_y(q) \leq \alpha_y^{\text{ex}}(x)$.*

Proof. $\mathcal{Z}(q)$ is the Minkowski sum of the segments $[T_{\min}, \bar{T}_i] q_i$, and the support function of a Minkowski sum is the sum of the support functions; for one segment, $\max_{T \in [T_{\min}, \bar{T}_i]} T q_i^\top y = \phi_i(q_i^\top y)$. Realizability follows from Assumption 4 with $u_i^\perp = 0$, since $y^\top \ddot{x}_L = \frac{1}{m_L} \sum_i T_i q_i^\top y - g e_3^\top y$ is affine in T on a box. $\square$

A. Systems with second-order authority

The construction uses two properties of the cable system, stated here for a general system with coordinates $\zeta_1, \dots, \zeta_N$: the braking authority is bounded above and below by nondecreasing functions of the coordinates, and each coordinate responds to the inputs only through its second derivative. For $\nu, \bar{\zeta} > 0$, the admissible swing set

$$\mathcal{V}(\nu, \bar{\zeta}) := \{(\zeta, \dot{\zeta}) \in \mathbb{R}^2 : -\bar{\zeta} + \frac{(\omega)^2}{2\nu} \leq \zeta \leq \bar{\zeta} - \frac{(\omega)^2}{2\nu}\} \quad (10)$$

is the convex set of $(\zeta, \dot{\zeta})$ from which a double integrator with $|\ddot{\zeta}| \leq \nu$ can keep $|\zeta| \leq \bar{\zeta}$ for all future time.

Definition 7 (Second-order authority). *Consider a system with admissible inputs, a constraint function h with $\dot{h} = -v$ and $\ddot{h} = -b$, a closed operational set $\mathcal{X}_{\text{op}}$, and twice continuously differentiable coordinates $\zeta_1, \dots, \zeta_N$ of the state. The system has second-order authority along h on $\mathcal{X}_{\text{op}}$ with data $(\underline{\alpha}, \bar{\alpha}, \nu, \nu^{\text{rel}}, \bar{\zeta})$ if:*

- (C1) $\underline{\alpha}$ is the sum of a constant and of N continuous, nondecreasing, piecewise-linear functions of the individual ζ_i ; $\bar{\alpha} \geq \underline{\alpha}$ is continuous and nondecreasing in each ζ_i ; and $b \leq \bar{\alpha}(\zeta)$ for every admissible input at every $x \in \mathcal{X}_{\text{op}}$.
- (C2) $|\zeta_i| \leq \bar{\zeta}$ on $\mathcal{X}_{\text{op}}$, and $|\ddot{\zeta}_i| \leq \nu^{\text{rel}}$ for every admissible input at every $x \in \mathcal{X}_{\text{op}}$.
- (C3) There is a map that assigns to every $x \in \mathcal{X}_{\text{op}}$ with $(\zeta_i, \dot{\zeta}_i) \in \mathcal{V}(\nu, \bar{\zeta})$ for all i , and to every choice of $\ddot{\zeta}_i^{\text{d}} \in [-\nu, \nu]$, an admissible input that produces $b = \bar{\alpha}(\zeta)$ and $\ddot{\zeta}_i = \ddot{\zeta}_i^{\text{d}}$ for all i , and whose solutions remain in $\mathcal{X}_{\text{op}}$ as long as $(\zeta_i, \dot{\zeta}_i) \in \mathcal{V}(\nu, \bar{\zeta})$ for all i .

In words, no admissible input brakes harder than $\bar{\alpha}$ or swings faster than ν^{rel} , while a reserved part of the input swings every coordinate at any rate up to ν and brakes with $\underline{\alpha}$ at the same time.

B. The braking maneuver and its stopping distance

For $(\zeta, \omega) \in \mathcal{V}(\nu, \bar{\zeta})$, let $\tilde{\zeta}(t; \zeta, \omega)$ be the fastest motion of $\tilde{\zeta} \in [-\nu, \nu]$ from (ζ, ω) to $(\bar{\zeta}, 0)$, held at $\bar{\zeta}$ afterwards: with $r := \sqrt{\omega^2/2 + \nu(\bar{\zeta} - \zeta)}$, $t_1 := (r - \omega)/\nu \geq 0$, and $t_2 := t_1 + r/\nu$,

$$\tilde{\zeta}(t) = \begin{cases} \zeta + \omega t + \frac{\nu}{2}t^2, & 0 \leq t \leq t_1, \\ \bar{\zeta} - \frac{\nu}{2}(t_2 - t)^2, & t_1 \leq t \leq t_2, \\ \bar{\zeta}, & t \geq t_2. \end{cases} \quad (11)$$

Let x have $(\zeta_i, \dot{\zeta}_i) \in \mathcal{V}(\nu, \bar{\zeta})$ for all i , and let every coordinate follow its swing $\tilde{\zeta}_i(t) := \tilde{\zeta}(t; \zeta_i, \dot{\zeta}_i)$. Along the maneuver, the braking deceleration is $\tilde{\alpha}(t; x) := \underline{\alpha}(\tilde{\zeta}_1(t), \dots, \tilde{\zeta}_N(t))$, the approach speed $\tilde{v}(t; x) := v - \int_0^t \tilde{\alpha}(s; x) ds$, and the distance traveled toward the wall $E(t; x) := \int_0^t \tilde{v}(s; x) ds$. The *stopping distance* is

$$D(x) := \max_{t \geq 0} E(t; x), \quad (12)$$

with maximizing times $T^*(x)$; the maximum exists if $\underline{\alpha}_{\text{sat}} := \underline{\alpha}(\bar{\zeta}, \dots, \bar{\zeta}) > 0$, and it is taken over all $t \geq 0$, not up to the first zero of $\tilde{v}$, because a negative authority accelerates a payload at rest toward the wall. The *authority barrier* is $H(x) := h - D(x)$, and

$$\mathcal{X}_{\text{RF}} := \{x \in \mathcal{X}_{\text{op}} : H(x) \geq 0, (\zeta_i, \dot{\zeta}_i) \in \mathcal{V}(\nu, \bar{\zeta}) \forall i\}. \quad (13)$$

$H(x)$ is the smallest distance to the wall that the payload reaches during the maneuver started at x . Let $\pi^\#(x)$ be the input that (C3) assigns to x with $\tilde{\zeta}_i^d = \tilde{\zeta}(0; \zeta_i, \dot{\zeta}_i)$; since the time-optimal swing is a feedback of $(\zeta_i, \dot{\zeta}_i)$, the closed loop under $\pi^\#$ reproduces the swings, $b(t) = \tilde{\alpha}(t; x_0)$, and $v(t) = \tilde{v}(t; x_0)$.

Lemma 8 (Envelope and monotonicity). (i) Every $\zeta(\cdot)$ with absolutely continuous derivative, $\zeta(0) = \zeta$, $\dot{\zeta}(0) = \omega$, $|\dot{\zeta}| \leq \nu$ a.e., and $\zeta \leq \bar{\zeta}$ satisfies $\zeta(t) \leq \tilde{\zeta}(t; \zeta, \omega)$ for all $t \geq 0$. (ii) $\tilde{\zeta}(t; \zeta, \omega)$ is nondecreasing in ζ and ω on $\mathcal{V}(\nu, \bar{\zeta})$. (iii) D is nondecreasing in v , nonincreasing in every ζ_i and $\dot{\zeta}_i$, and locally Lipschitz; along the maneuver started at x_0 , $D(x(t)) = \max_{s \geq t} \tilde{v}(\cdot; x_0) - \int_0^t \tilde{v}(\cdot; x_0)$.

Proof scheme. Write $S(\zeta) = \sqrt{2\nu(\bar{\zeta} - \zeta)}$. A feasible trajectory moving toward $\bar{\zeta}$ must satisfy $\dot{\zeta} \leq S(\zeta)$, or even full braking would overshoot. Before the switch t_1 , $\tilde{\zeta}$ uses the largest acceleration ν ; afterwards it follows $\dot{\zeta} = S(\zeta)$ to $\bar{\zeta}$. A first crossing from below either contradicts maximal acceleration before t_1 or forces both motions to coincide on the stopping curve after t_1 . This proves (i). Differentiation of the two polynomial branches in (11) gives nonnegative initial-state sensitivities on $\mathcal{V}$, proving (ii). Monotonicity of $\underline{\alpha}$ then orders $E(t)$ for every t , and maximization preserves that order. On compact initial-state sets the swing arrival times and the maximizing times are uniformly bounded because $\underline{\alpha}_{\text{sat}} > 0$;

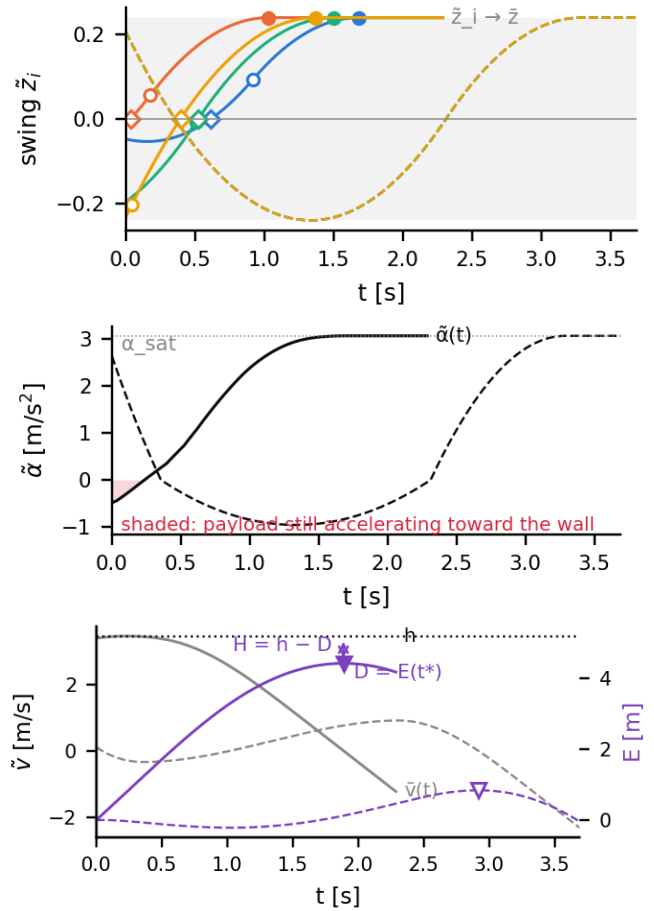


Fig. 2. A reserved swing (top) changes the cable directions and increases available braking (middle). The barrier subtracts the maximum wallward excursion $\max_t E(t)$ from the current distance h (bottom). Solid and dashed curves are two example states.

taking the maximum of the resulting locally Lipschitz family proves (iii). Restarting the swing at time t reproduces its remaining trajectory, giving the stated time-shift identity. $\square$

C. Main result

At regular swing states, for fixed t , $E(t; x)$ is continuously differentiable in $(v, \zeta, \dot{\zeta})$, because (11) is continuously differentiable in its initial data (one-sided at $(\bar{\zeta}, 0)$) and the integration over s smooths the breakpoints of $\underline{\alpha}$, which each $\tilde{\zeta}_i$ meets only at isolated times except while held at $\bar{\zeta}$. At these regular states, Danskin's theorem [12] gives the directional derivative

$$D'(x; \delta) = \max_{t \in T^*(x)} \nabla_x E(t; x)^\top \delta \quad (14)$$

in directions δ along which the state stays in the domain of D . At a degenerate swing state, use the upper right Dini derivative instead of the displayed gradient formula. For an admissible u , let $F_u(x)$ be the state derivative and define $H'(x; F_u(x))$ as this Dini derivative; at regular states it equals $-v - D'(x; F_u(x))$, the ordinary derivative $\dot{H}(x, u)$ when the maximizing time is unique. On states with all $(\zeta_i, \dot{\zeta}_i) \in \mathcal{V}(\nu^{\text{rel}}, \bar{\zeta})$, let D^{rel} be (12) computed with $(\bar{\alpha}, \nu^{\text{rel}})$ in place of $(\underline{\alpha}, \nu)$, the stopping distance of an optimistic system that

swings and brakes as hard as any admissible input allows, and let γ be a locally Lipschitz extended class- $\mathcal{K}$ function.

Theorem 9 (Inner and outer bounds). *Let the system have second-order authority along h on $\mathcal{X}_{\text{op}}$ with data $(\underline{\alpha}, \bar{\alpha}, \nu, \nu^{\text{rel}}, \bar{\zeta})$, and let $\underline{\alpha}_{\text{sat}} > 0$. Then:*

- (i) $\mathcal{X}_{\text{RF}}$ is controlled invariant. For every $x \in \mathcal{X}_{\text{RF}}$, $K_H(x) := \{u \text{ admissible} : H'(x; F_u(x)) \geq -\gamma(H(x))\}$ contains $\pi^\sharp(x)$. Every solution generated by a measurable selection $u(t) \in K_H(x(t))$ from $x(0) \in \mathcal{X}_{\text{RF}}$ that keeps $(\zeta_i, \dot{\zeta}_i) \in \mathcal{V}(\nu, \bar{\zeta})$ satisfies $h(t) \geq H(x(t)) \geq 0$ for all $t \geq 0$.
- (ii) $\text{Viab}(\mathcal{C}) \subseteq \{x \in \mathcal{X}_{\text{op}} : (\zeta_i, \dot{\zeta}_i) \in \mathcal{V}(\nu^{\text{rel}}, \bar{\zeta}) \forall i, h \geq D^{\text{rel}}(x)\}$.
- (iii) If $\underline{\alpha} \equiv \bar{\alpha} > 0$, then $D = (v)_+^2 / (2\bar{\alpha})$ and $\mathcal{X}_{\text{RF}}$ is the classical safe set of a double integrator.

Proof. (i) Under $\pi^\sharp$ from $x_0 \in \mathcal{X}_{\text{RF}}$, Lemma 8(iii) and $h(t) = h_0 - \int_0^t v$ give $H(x(t)) = h_0 - \max_{s \geq t} \int_0^s v$, nondecreasing in t ; $\pi^\sharp$ keeps the state in $\mathcal{X}_{\text{op}}$ and the swing states in $\mathcal{V}$, so $\mathcal{X}_{\text{RF}}$ is forward invariant under $\pi^\sharp$, hence controlled invariant. Next, the swing restarted at a later state of the maneuver is the remaining part of the swing started at x , so the derivative of $E(t; \cdot)$ along $F_{\pi^\sharp}(x)$ equals $\tilde{v}(t; x) - v$ at regular states, and by (14), $H'(x; F_{\pi^\sharp}(x)) = -\max_{t \in T^*(x)} \tilde{v}(t; x)$. Every maximizing time $t > 0$ is a stationary point of $E(\cdot; x)$, so $\tilde{v}(t; x) = 0$, and $t = 0$ maximizes only if $v \leq 0$; hence $H'(x; F_{\pi^\sharp}(x)) \geq 0 \geq -\gamma(H(x))$ at regular states. The preceding time-shift identity gives the same inequality for the Dini derivative along $\pi^\sharp$ at degenerate states. Finally, along a solution generated by K_H that keeps the swing states in $\mathcal{V}$, D is locally Lipschitz [12], so $H(x(t))$ is continuous with derivative $H'(x; \dot{x}) \geq -\gamma(H)$ a.e., and the comparison argument for nonsmooth barriers [13] gives $H \geq 0$; $h \geq H$ since $D \geq 0$.

(ii) Let $x_0 \in \text{Viab}(\mathcal{C})$ and let an admissible input keep the solution in $\mathcal{C}$. By (C2), $(\zeta_i, \dot{\zeta}_i) \in \mathcal{V}(\nu^{\text{rel}}, \bar{\zeta})$ at all times, and by Lemma 8(i) with ν^{rel} , $\zeta_i(t) \leq \bar{\zeta}_i^{\text{rel}}(t)$. By (C1) and the monotonicity of $\bar{\alpha}$, $b(t) \leq \bar{\alpha}(\zeta(t)) \leq \bar{\alpha}(\bar{\zeta}^{\text{rel}}(t)) = \bar{\alpha}^{\text{rel}}(t; x_0)$, so $v(t) \geq \tilde{v}^{\text{rel}}(t; x_0)$ and $\min_t h(t) \leq h_0 - D^{\text{rel}}(x_0)$; if $h_0 < D^{\text{rel}}(x_0)$, the solution leaves $\mathcal{C}$.

(iii) $\bar{\alpha} \equiv \bar{\alpha}$ gives $\tilde{v} = v - \bar{\alpha}t$ and $\max_t \int_0^t \tilde{v} = (v)_+^2 / (2\bar{\alpha})$. $\square$

Together, (i) and (ii) give $\mathcal{X}_{\text{RF}} \subseteq \text{Viab}(\mathcal{C}) \subseteq \{h \geq D^{\text{rel}}\}$, which answers problems (a) and (b) with $\mathcal{S} = \mathcal{X}_{\text{RF}}$; Sec. V answers (c). The gap $D - D^{\text{rel}}$ is what the division of the input costs and vanishes in case (iii). Part (i) is a backup-CBF guarantee [3], [14] with $\pi^\sharp$ as the backup controller, but the closest approach along the backup trajectory has the closed form (12) instead of being simulated, and the same data yield the outer bound (ii). The sets $\mathcal{X}_{\text{RF}}$ and $\mathcal{C}_{\text{HO}}$ need not be nested.

D. Application to the cable system

Assumption 10 (Swing reserve). *There exist $\nu, \nu_w > 0$ and $\rho_i^{\text{fb}} \geq 0$ such that, for every i , $\sec \theta_q \sqrt{\bar{\eta}_{z,i}^2 + \bar{\eta}_{w,i}^2} + \rho_i^{\text{fb}} \leq \rho_i$,*

where $\bar{\eta}_{z,i} := m_i(l_i \nu + a_{\text{max}} + l_i \bar{\omega}^2 \bar{z})$ and $\bar{\eta}_{w,i} := m_i(l_i \nu_w + a_{\text{max}} + l_i \bar{\omega}^2 \bar{w})$.

Assumption 11 (Swing-rate consistency). $\bar{\omega}^2 \geq \sec^2 \theta_q (4\nu \bar{z} + 4\nu_w \bar{w})$.

Assumption 10 reserves, within ρ_i , the perpendicular thrust needed to impose any $\ddot{z}_i \in [-\nu, \nu]$ and $\ddot{w}_i \in [-\nu_w, \nu_w]$ against the coupling terms of Lemma 1(b), the factor $\sec \theta_q$ accounting for the non-orthogonality of the two swing directions on the sphere, and keeps ρ_i^{fb} for feedback; Assumption 11 makes the swing-rate bound of (4) hold on the admissible swing sets.

Proposition 12 (The cable system has second-order authority). *Let Assumptions 4, 10, and 11 hold, let $d_L = d_i = 0$, and let $\zeta_i := z_i$. Then (1)–(2) has second-order authority along h on $\mathcal{X}_{\text{op}}$ with $\underline{\alpha} = \alpha_y$ of (9),*

$$\bar{\alpha}(z) = \min \left\{ \frac{1}{m_L} \sum_i \phi_i^{\text{rel}}(z_i), \right. \\ \left. a_{\text{max}} \sec \theta_q \max_i (z_i)_+ \right\}, \quad (15)$$

where ϕ_i^{rel} is ϕ_i with $\bar{T}_i^{\text{rel}} := f_{\text{max},i} + m_i a_{\text{max}} + m_i l_i \bar{\omega}^2$; ν of Assumption 10; $\nu^{\text{rel}} := \max_i (\frac{f_{\text{max},i}}{m_i l_i} + \frac{a_{\text{max}}}{l_i} + \bar{\omega}^2)$; and $\bar{\zeta} = \bar{z}$. The map of (C3) sets $T_i = \bar{T}_i$ if $z_i > 0$ and $T_i = T_{\text{min}}$ otherwise, and steers each w_i within $\mathcal{V}(\nu_w, \bar{w})$.

Proof scheme. For (C1), $T_i \leq f_{\text{max},i} + m_i a_{\text{max}} + m_i l_i \bar{\omega}^2$ follows from Lemma 1(a). The tension cone also gives $\sum_i T_i \cos \theta_q \leq m_L a_{\text{max}}$, which bounds b by the second term of (15). Projecting Lemma 1(b) onto y establishes (C2). For (C3), choose the tensions that realize $\underline{\alpha}$ and prescribe $\ddot{z}_i \in [-\nu, \nu]$ plus a transverse $\ddot{w}_i \in [-\nu_w, \nu_w]$ that keeps w_i in its stopping set. The Gram matrix of $(P_i y, P_i r)$ has eigenvalues 1 and $1 - z_i^2 - w_i^2 \geq \cos^2 \theta_q$, so the required perpendicular thrust has norm at most $\sec \theta_q \sqrt{\bar{\eta}_{z,i}^2 + \bar{\eta}_{w,i}^2} \leq \rho_i$. Assumption 4 then realizes the tensions and swings simultaneously. Finally, $\|\dot{q}_i\|^2 \leq \sec^2 \theta_q (\dot{z}_i^2 + \dot{w}_i^2) \leq \bar{\omega}^2$ on the product of the swing stopping sets, proving that the trajectory stays in $\mathcal{X}_{\text{op}}$. $\square$

The second term of (15) comes from the cone of (5): since $\sum_i T_i \cos \theta_q \leq e_3^\top \sum_i T_i q_i \leq m_L a_{\text{max}}$, one has $b = \frac{1}{m_L} \sum_i T_i z_i \leq a_{\text{max}} \sec \theta_q \max_i (z_i)_+$. With all cables at $\bar{z}$, $D^{\text{rel}} = (v)_+^2 / (2a_{\text{max}} \bar{z} \sec \theta_q)$ against $D = (v)_+^2 / (2\underline{\alpha}_{\text{sat}})$ with $\underline{\alpha}_{\text{sat}} = \frac{1}{m_L} \sum_i \bar{T}_i \bar{z}$; where the cables must first swing, the gap is set by ν against ν^{rel} .

Proposition 13 (Computation and decomposition). *For $t_j \in T^*(x)$, the time derivative of $h - E(t_j; \cdot)$ under u is*

$$\dot{H}_j(x, u) = -v + t_j b \\ - \sum_i (\partial_{z_i} E(t_j; x) \dot{z}_i + \partial_{w_i} E(t_j; x) \dot{w}_i), \quad (16)$$

with $b = \frac{1}{m_L} \sum_i T_i z_i$ and $\ddot{z}_i = \frac{1}{m_i l_i} y^\top u_i^\perp - \frac{1}{l_i} y^\top P_i a - \|\dot{q}_i\|^2 z_i$, and $H'(x; F_u(x)) = \min_j \dot{H}_j(x, u)$ at regular states. Moreover: (a) D , T^* , and the coefficients of (16) depend on the state only through $(v, \{z_i, \dot{z}_i\}_i)$ and are computed exactly in $O(N \log N)$ arithmetic operations, because $E(\cdot; x)$

is piecewise quartic with at most $4N$ breakpoints; forming all gradient rows additionally takes $O(N|T^*|)$ time; (b) given a common value $\hat{a}$ of the specific force, each constraint $\dot{H}_j \geq -\gamma(H)$ splits into N affine constraints, one per quadrotor on $(T_i, u_i^\perp)$, whose right-hand sides sum to $-\gamma(H) + v$ once the consistency equation $m_L \hat{a} = \sum_i T_i q_i$ is also enforced; (c) without a common $\hat{a}$, the local constraints are neither necessary nor sufficient.

Proof scheme. At a regular state, $\partial_v E(t; x) = t$ and the chain rule gives (16); Danskin's formula makes the derivative of $H = h - \max_t E(t)$ the minimum over its active times. Every $\tilde{z}_i$ is piecewise quadratic, with at most its switch, arrival, and two zero crossings as breakpoints. Thus $\tilde{\alpha}$ is piecewise quadratic, E piecewise quartic, and candidate maximizers are interval endpoints or roots of cubic $\tilde{v}$. Sorting $O(N)$ events takes $O(N \log N)$ arithmetic operations; all active gradient rows additionally take $O(N|T^*|)$. For a committed common $\hat{a}$, (16) is a sum of local affine terms, but validity also requires $m_L \hat{a} = \sum_i T_i q_i$; otherwise the local rows need not represent the actual coupled plant. $\square$

Remark 14 (Scope of the guarantee). The certificate assumes an exact taut-cable model, continuously updated thrust commands, and a feasible transverse swing reserve. Wind, mass errors, actuator tracking, slack cables, and zero-order-hold inputs require additional error bounds and are *not* covered by Theorem 9. In particular, continuity and local Lipschitzness of D do not by themselves give a numerical sampled-data margin.

V. SAFETY FILTER

The decision variables are the tensions $T \in \mathbb{R}^N$ and the perpendicular components $u^\perp = (u_1^\perp, \dots, u_N^\perp)$, $u_i^\perp \perp q_i$; the thrust vector of quadrotor i is $s_i(T)q_i + u_i^\perp$ with $a = \frac{1}{m_L} \sum_j T_j q_j$ in Lemma 1(a). For each swing coordinate $\zeta \in \{z_i, w_i\}$ with data $(\nu_\zeta, \bar{\zeta}_\zeta) \in \{(\nu, \bar{z}), (\nu_w, \bar{w})\}$, let $\partial_\zeta^\pm$ be the upper and lower boundaries of $\mathcal{V}(\nu_\zeta, \bar{\zeta}_\zeta)$. Given the nominal thrust vectors u_i^{nom} , the filter solves

$$\min_{T, u^\perp} \sum_i \|s_i(T)q_i + u_i^\perp - u_i^{\text{nom}}\|^2 \quad (17a)$$

$$\text{s.t. } u \in \mathcal{U}(x), \quad (17b)$$

$$\dot{H}_j(x, T, u^\perp) \geq -\gamma(H(x)) \quad \forall t_j \in T^*(x), \quad (17c)$$

$$\dot{\zeta} \leq -\nu_\zeta \mathbf{1}[\dot{\zeta} > 0] \text{ on } \partial_\zeta^+, \quad \dot{\zeta} \geq \nu_\zeta \mathbf{1}[\dot{\zeta} < 0] \text{ on } \partial_\zeta^-, \quad (17d)$$

for every i and every swing coordinate. Constraint (17c) is the barrier condition of Thm. 9(i) written with Prop. 13, one affine constraint per maximizing time, and (17d) are the tangency conditions that are necessary for the swing states to remain in their admissible sets [9]; a class- $\mathcal{K}$ condition on a function that vanishes on $\partial\mathcal{V}$ would forbid the time-optimal swing, which reaches that boundary in finite time. At regular states the program is a second-order cone program with $3N$ variables, with global coupling through the common payload force a , the thrust reconstructions $s_i(T)$, and the cone in (5). An affine split of the barrier rows still requires enforcing the common-force consistency equation. At degenerate swing states, the

displayed gradient rows are not asserted to characterize the Dini condition; the filter applies the known feasible backup input $\pi^\sharp$ instead.

By Thm. 9(i), the filter is feasible at regular states of $\mathcal{X}_{\text{RF}}$ and applies $\pi^\sharp$ at degenerate states; at regular states $\pi^\sharp(x)$ satisfies (17c) with slack at least $\gamma(H)$; along every solution whose swing states remain admissible, $h \geq H \geq 0$.

In a distributed implementation, the broadcast $\hat{a}$ must equal the specific force generated by the simultaneous tension commands, and the cone and thrust constraints still require coordination. A lagged broadcast is not a feasibility certificate.

Uncertain mass requires checking the coupled thrust, tension, and swing reserve for every allowed mass; replacing m_L only in the authority formula is insufficient for a robust guarantee.

VI. NUMERICAL STUDY

Figures 2 and 6–8 illustrate the predicted and realized swing profiles. The full-order simulation has four 1.5 kg vehicles carrying a 1 kg point payload on 1 m cables. It uses $T_{\min} = 1$ N, $f_{\max} = 44$ N, $\bar{T}_i = 3.2$ N, $\rho_i = 34$ N, $a_{\max} = 13$ m/s², $\bar{z} = 0.239$, $\bar{w} = 0.171$, $\bar{\omega} = 1$ s⁻¹, and $\nu = \nu_w = 0.5$ s⁻². Figure 11 shows the thrust split. The thrust-reserve bound is arithmetically consistent: $\sqrt{(\bar{T}_i + m_i a_{\max} + m_i l_i \bar{\omega}^2)^2 + \rho_i^2} = 41.73$ N $< (f_{\max} = 44$ N). Its spring-damper cables can go slack; the filter runs at 200 Hz. The code of the simulations and videos of the trials are available online.¹

All 690 HOCBF trials over the tested gain pairs and actuator modes lost QP feasibility before a cable entered the braking half-space (Fig. 4); this does not establish initial feasibility for arbitrary gains. Of 100 trials started close to $H = 0$ under an adversarial nominal input, the sampled authority filter reached the wall in 59 despite a feasible QP at its sampling instants (Fig. 5). In 56 trials the spring-damper cables temporarily lost the positive-tension condition. Thus the continuous-time invariant-set theorem does not certify these runs. The largest barrier loss was 0.85 m within one 5 ms hold; no contact was observed above an initial margin of 0.78 m in the tested 100-state sweep. This margin is empirical and holds for the tested distribution only. A 1 kHz attitude loop also incurred 17–43 N tracking errors during command jumps.

Direct-collocation searches covered three initial cable configurations at eight speeds, with each candidate subsequently simulated. When cables already lean away from the wall, the observed feasible distance is about 0.97–0.98 times D ; when a swing is required, feasible trajectories can begin at only 0.25–0.55 D (Figs. 9 and 10). Figure 3 shows one of the found trajectories. The large gap when cables lean toward the wall is driven in part by the reserved swing acceleration: $\nu = 0.5$ s⁻² needs about 1.8 s to cross the allowed angle range, while the physical thrust in the simulations can reach the swing-rate limit much sooner. Figure 12 varies a hypothetical ν to show this sensitivity; values beyond the shaded admissible region

¹https://github.com/Hadi-Hajieghrary/Authority_Barriers

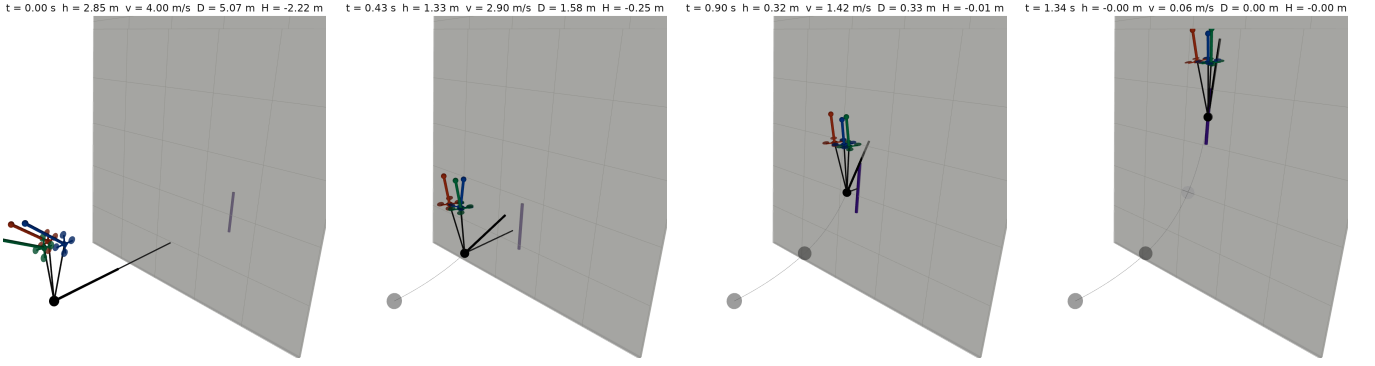


Fig. 3. Four frames of a collocation trajectory for the mixed cable configuration at $v_0 = 4$ m/s from $h_0 = 0.56D$. The vehicles reposition their cables behind the payload and enter the certified set. The frames show one example of a safe trajectory; the kernel was not computed exhaustively.

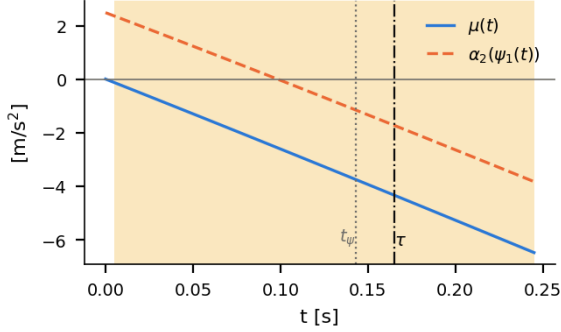


Fig. 4. HOCBF feasibility margin in a perfect-actuator trial. The feasible-to-infeasible transition occurs before a cable can enter the braking half-space.

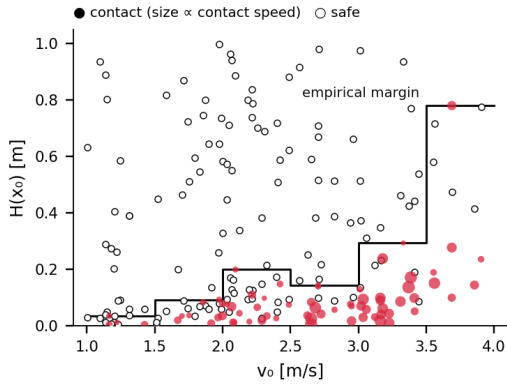


Fig. 5. Sampled-filter runs in the $(v_0, H(x_0))$ plane (filled: wall contact; open: no contact). The step gives the largest observed contact margin in each speed bin. The observed threshold is not a certified sampled-data bound.

cannot be substituted into the theorem without changing its assumptions. A collocation success gives an example trajectory but a failure does not locate the viability boundary. The numerical evidence therefore supports the qualitative gap between the reserved and relaxed models, while it cannot validate either set inclusion or a universal percentage gap.

Evaluating D takes 0.5–0.6 ms and the filter 2–4.5 ms, close to the 5 ms sampling period; solver failures, model mismatch, and inter-sample excursions preclude a real-time guarantee. A certified zero-order-hold error bound and an explicit rate constraint are needed before claims of sampled safety are made.

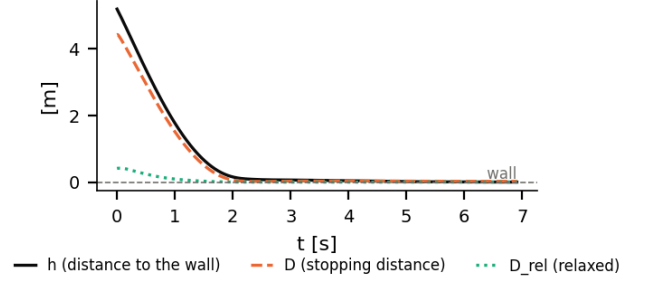


Fig. 6. Wall distance h (solid), predicted stopping distance D (dashed), and optimistic stopping distance D^{rel} (dotted) in a perfect-actuator trial under an adversarial nominal command. The trial illustrates the continuous-time maneuver but does not prove sampled safety.

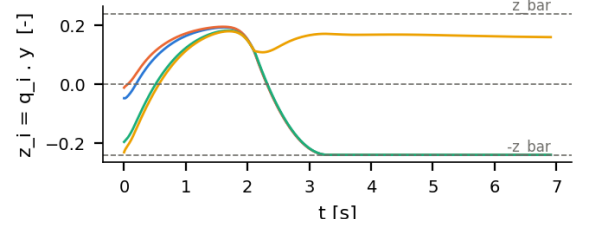


Fig. 7. Cable swing coordinates z_i of the four vehicles in the trial of Fig. 6.

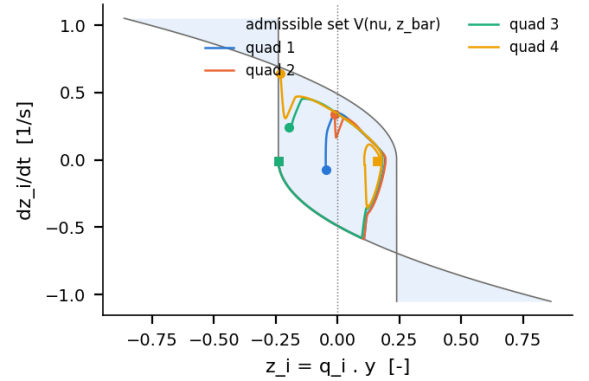


Fig. 8. Swing states $(z_i, \dot{z}_i)$ in the trial of Fig. 6 (circle: start; square: end); the shaded region is the admissible swing set $\mathcal{V}$.

VII. CONCLUSION

Cable geometry makes braking authority a controlled state: the team brakes only as hard as the cable directions allow,

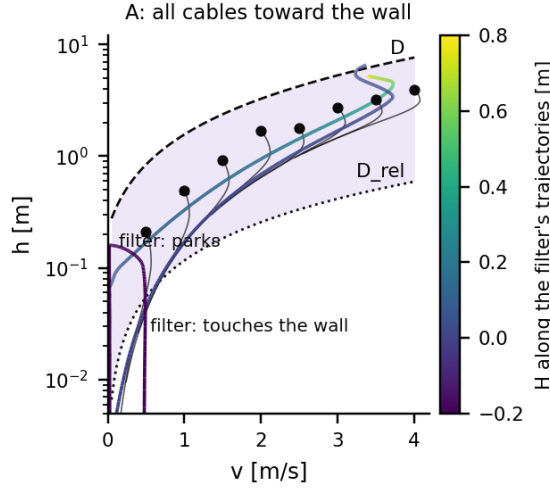


Fig. 9. Viability band for configuration A, cables initially leaning toward the wall: reserved (dashed) and optimistic (dotted) stopping distances. Dots are the smallest *found* and *re-simulated* collocation trajectories; they do not prove where the kernel boundary lies. Colored curves are two closed-loop trajectories of the sampled filter.

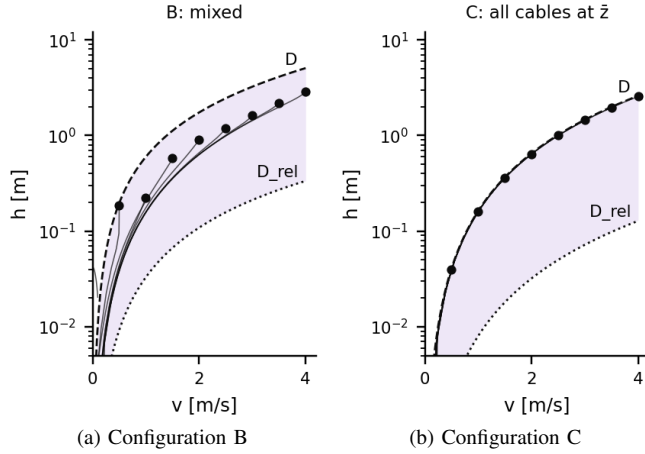


Fig. 10. Viability bands for (a) mixed cable directions and (b) cables already behind the payload; curves and dots as in Fig. 9.

and these change at a bounded rate. An initially feasible HOCBF can lose feasibility when the geometry cannot change quickly enough, under the stated gain-dependent condition. A support-function calculation and an explicit swing-and-brake maneuver give useful continuous-time inner and outer viability bounds, and the argument extends to every system with second-order authority. The inner bound is conservative where the cables must first swing: safe trajectories were found from as little as $0.25 D$ there, against 0.97 – $0.98 D$ with the cables already behind the payload; the reserved swing acceleration accounts for part of this gap.

The simulations also reveal that the current sampled filter and physical cable model do not satisfy the theorem's hypotheses: the barrier lost up to 0.85 m within one 5 ms hold; of 100 trials started near the boundary of the safe set, 56 had a slack cable and 59 reached the wall. A certified sampled-data bound, coordinated force allocation, and reproducible full-order validation are the next requirements for a deployment claim.

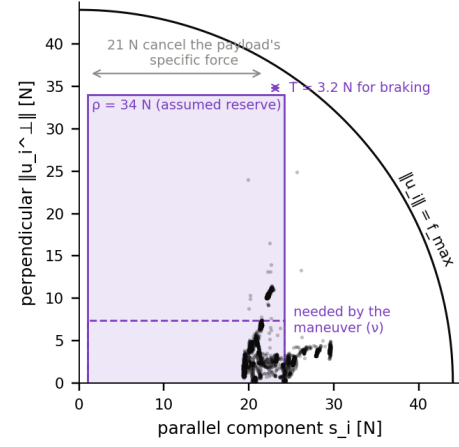


Fig. 11. Thrust budget of one vehicle: the reserved rectangle lies in the available thrust ball.

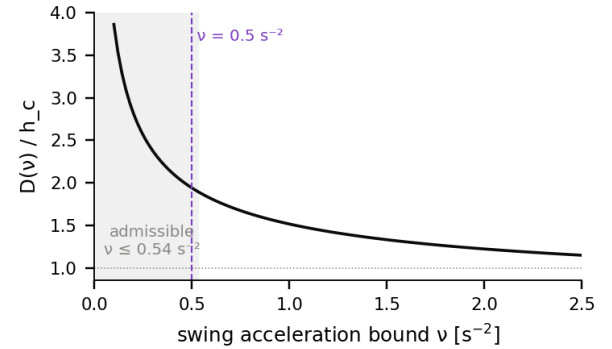


Fig. 12. Sensitivity of the reserved stopping distance to a hypothetical swing acceleration ν for configuration A at $v_0 = 3$ m/s. The shaded interval respects the stated rate-consistency assumption; points to its right are diagnostic calculations, not certified operating points.

REFERENCES

- [1] W. Xiao and C. Belta, "High-order control barrier functions," *IEEE Trans. Autom. Control*, vol. 67, no. 7, pp. 3655–3662, 2022.
- [2] J. Breeden and D. Panagou, "High relative degree control barrier functions under input constraints," in *Proc. IEEE Conf. Decision Control (CDC)*, 2021, pp. 6119–6124.
- [3] Y. Chen, M. Jankovic, M. A. Santillo, and A. D. Ames, "Backup control barrier functions: Formulation and comparative study," in *Proc. IEEE Conf. Decision Control (CDC)*, 2021, pp. 6835–6841.
- [4] D. E. J. van Wijk, E. Daş, T. G. Molnár, A. D. Ames, and J. W. Burdick, "Safety-critical control with bounded inputs: A closed-form solution for backup control barrier functions," in *Proc. American Control Conf. (ACC)*, 2026, pp. 1215–1222.
- [5] M. Gouttefarde, D. Daney, and J.-P. Merlet, "Interval-analysis-based determination of the wrench-feasible workspace of parallel cable-driven robots," *IEEE Trans. Robot.*, vol. 27, no. 1, pp. 1–13, 2011.
- [6] J. C. Erskine, A. Chriette, and S. Caro, "Wrench analysis of cable-suspended parallel robots actuated by quadrotor unmanned aerial vehicles," *J. Mechanisms Robot.*, vol. 11, no. 2, p. 020909, 2019.
- [7] K. Sreenath and V. Kumar, "Dynamics, control and planning for cooperative manipulation of payloads suspended by cables from multiple quadrotor robots," in *Proc. Robotics: Science and Systems*, Berlin, Germany, June 2013.
- [8] T. Lee, "Geometric control of quadrotor UAVs transporting a cable-suspended rigid body," *IEEE Trans. Control Syst. Technol.*, vol. 26, no. 1, pp. 255–264, 2018.
- [9] J.-P. Aubin, *Viability Theory*. Boston, MA: Birkhäuser Boston, 1991.
- [10] W. Xiao and C. Belta, "Control barrier functions for systems with high relative degree," in *Proc. IEEE Conf. Decision Control (CDC)*, 2019, pp. 474–479.

- [11] Q. Nguyen and K. Sreenath, "Exponential control barrier functions for enforcing high relative-degree safety-critical constraints," in *Proc. American Control Conf. (ACC)*, 2016, pp. 322–328.
- [12] F. H. Clarke, *Optimization and Nonsmooth Analysis*, ser. Classics in Applied Mathematics. SIAM, 1990, vol. 5.
- [13] P. Glotfelter, J. Cortés, and M. Egerstedt, "Nonsmooth barrier functions with applications to multi-robot systems," *IEEE Control Syst. Lett.*, vol. 1, no. 2, pp. 310–315, 2017.
- [14] T. Gurriet, M. Mote, A. Singletary, P. Nilsson, E. Feron, and A. D. Ames, "A scalable safety critical control framework for nonlinear systems," *IEEE Access*, vol. 8, pp. 187 249–187 275, 2020.